

# Sieve: Budgeted Question-Conditioned Information Selection for Training-Free Long-Context Compression

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## Abstract

We formulate long-context compression as *budgeted question-conditioned information selection*: given a fixed word budget, choose the sentence subset that jointly serves question-relevance coverage, positional reliability, and diversity, with the selector’s own cost a first-class constraint. Sieve instantiates this formulation without any hosted model: IDF-weighted coverage of query content terms, a U-shaped positional prior, and a redundancy penalty applied greedily under the budget. Preprocessing is linear-time, greedy selection is  $O(k n \bar{s})$ , and the selector runs in tens of milliseconds on one CPU core. On three evidence-annotated benchmarks (QASPER, HotpotQA, 2WikiMultihopQA; 1600 records), three protocols measure the predicted trade-offs: a reader-free evidence-recall protocol, end-to-end question answering with a production reader (seven methods,  $n=100$  each), and a full configuration–budget–dataset matrix. Question conditioning over truncation buys 12.8, 44.8, and 22.8 recall points at a  $4\times$  budget. Relevance-only selection, whether BM25 or the internal relevance term, is a reference ceiling for evidence recovery; the positional prior and redundancy penalty are objectives that spend recall—a dataset-stable six-to-seven paired points for the prior— at no additional compute. End to end the same ordering appears: BM25 retains 76.0% of full-context F1 against 61.8% for Sieve (paired gap  $|-4.8|$ , interval excluding zero), while Sieve remains statistically matched with truncation. A measured GPT-2 perplexity-gate configuration, the cost floor of the neural family, does not exceed the statistical selector on this benchmark at  $320\text{--}255\times$  the measured cost. The result is a formulation, a measured three-axis frontier, and a no-hosted-model reference point against which any context compressor can be priced.

**Keywords:** prompt compression, long-context question answering, information selection, efficiency, retrieval

## 1 Introduction

Long context is now a headline capability of large language models, and it is billed by the token. Serving a 32k-token prompt costs the customer proportionally more, costs the provider a prefill pass whose FLOPs scale linearly in prompt length, and occupies key-value cache memory that scales with both length and model depth. In document question answering, retrieval-augmented generation, and agent pipelines, the context that surrounds a query is often one or two orders of magnitude longer than the few sentences that actually determine the answer. Compressing the context before submission attacks all three costs at once.

The selector landscape is best read as a two-axis design space. The first axis is *selection-objective richness*: does a selector consult the question at all, does it account for where evidence sits, and does it control redundancy. The second axis is *infrastructure*: what must be hosted to run it. The last three years have filled the rich-objective, model-dependent corner quickly: LLMingua and its successors gate tokens by the perplexity of a hosted small language model (Jiang et al. 2023, 2024; Pan et al. 2024); Selective Context filters by self-information under a language model (Li et al. 2023); Perception Compressor, though training-free, retrieves demonstrations with Sentence-BERT and performs token-level compression with LLaMA-2-7B (Tang et al. 2025); QASPC is training-free, question-aware, and sentence-level (Chen et al. 2025); and a 2025 survey now maps the family systematically (Li et al. 2025). In the usage that has consolidated across this literature, “training-free” means free of *training*—not free of *models*: every method above pays for at least one hosted scorer, a tokenizer matched to it, and GPU time for forward passes over the full context. The opposite corner is occupied by the classical model-free selectors—BM25, TextRank, maximal marginal relevance (Robertson and Zaragoza 2009; Mihalcea and Tarau 2004; Carbonell and Goldstein 1998)—which run anywhere but, with one exception, ignore the question, the position, or the redundancy structure of the context; the exception, question-aware BM25, treats relevance as the whole objective. To our knowledge—and we audit this against a thirteen-entry design-space comparison (Table 1)—the region that combines richer selection objectives with no hosted model remains unoccupied: the closest published relatives either host a scorer (LongLLMingua, Perception Compressor, and, as stated in its abstract, QASPC), operate model-light graph machinery without positional or budget axes (Prompt-SAW (Ali et al. 2024)), or stop at relevance ranking (BM25). This paper formulates the selection problem that region calls for, instantiates it, and measures what the region can and cannot deliver on a quality–compression–cost frontier.

A natural objection precedes the study: why build a selector at all when classical retrieval already ranks sentences by query matching, and simply retrieving the top- $k$  relevant sentences is the natural baseline? The objection is correct as far as it goes, and we therefore implement its strongest form, BM25 sentence ranking under the same

word budget, as a baseline throughout. The answer the measurements give is that relevance ranking is the right *term* of a selector but not the whole of one. Raw relevance maximizes evidence recall (BM25 retains more gold evidence than our full selector on all three benchmarks), but recall is not the only quantity that matters to a long-context reader: where the retained evidence sits also determines whether the reader uses it, and how much of the budget survives near-duplicate sentences also determines how much evidence the budget can carry. Sieve is therefore an instance of *budgeted information selection*: an IDF-weighted question-relevance term, a positional prior that spends the budget where readers attend reliably, and a redundancy penalty under an explicit word budget, composed by a greedy fill. The formulation is the paper’s primary contribution and Sieve its no-hosted-model instantiation; the experiments then price each term separately, and the frontier picture that results is the second contribution: from truncation to question conditioning the quality gain is nearly free; from relevance ranking to the full selector, quality in the reader-free metric is *traded* for placement—a structural trade-off between selection objectives that relevance-only selectors cannot reach, not a deficiency of either family; and at the gate’s floor configuration, the measured GPT-2 scorer spends three orders of magnitude more compute than Sieve without exceeding it on this benchmark.

Our vehicle is Sieve, a sentence-level selector built from three statistics that are cheap, corpus-local, and question-directed. A sentence scores higher when it covers query terms with high inverse document frequency computed over the context itself, when it sits near the beginning or end of the context where models attend most reliably, and when it is dissimilar to sentences already selected; the last term is a redundancy penalty applied greedily under a word budget, which makes the whole rule an instance of maximal marginal relevance (Carbonell and Goldstein 1998) specialized with IDF weighting (Robertson and Zaragoza 2009) and a positional prior. Selection preserves original sentence order, so the compressed context remains coherent discourse rather than a bag of fragments. The entire pipeline is a few hundred lines of Python with no dependencies beyond the standard library and NumPy.

This article extends our earlier conference study of Sieve in directions the conference format could not accommodate. First, we reframe the evaluation around a three-way trade-off between task quality, compression ratio, and the computational cost of the compressor itself, and we report all three axes for every method, including a neural perplexity gate measured on the same hardware as Sieve (section 9). Second, we widen the evidence base from one benchmark to three: QASPER question answering over scientific papers (Dasigi et al. 2021), HotpotQA distractor multi-hop QA (Yang et al. 2018), and 2WikiMultihopQA compositional multi-hop QA (Ho et al. 2020), 1600 evidence-annotated records in total under a reader-free recall protocol, which answers the question the single-dataset study could not: whether question-conditioned selection generalizes beyond the domain it was designed on. Third, we strengthen the baseline set with BM25 sentence ranking, a strong question-conditioned model-free comparator, which the conference version lacked entirely. Fourth, we add a formal layer: a problem formulation that makes the trade-off explicit (section 3), propositions that bound the selection behavior under a stated lexical-coverage condition and the cost of each pipeline stage (section 5), and a robustness analysis covering evidence

placement, hyperparameter sensitivity, and cross-dataset transfer without retuning (section 10), together with a sample-level correlation study that quantifies how loosely evidence recall bounds end-task answer accuracy (section 11).

The empirical picture that emerges is sharper than the conference version could show. On the recall protocol, Sieve beats every question-agnostic selector on all three datasets with fully separated paired intervals, and the margin over truncation is largest exactly where the question’s lexical signal is strongest: multi-hop questions that name their entities. At  $4\times$  compression the margin over head truncation is 12.8 recall points on QASPER, 44.8 on HotpotQA, and 22.8 on 2WikiMultihopQA. BM25 sharpens the same picture from the other side: it retains 43.1% against Sieve’s 36.2% on QASPER, and the same ordering holds on the other two pools, tracking the selector’s own relevance-only ablation within a point on every dataset; the difference is exactly the price the positional prior pays for placing evidence where readers use it, and the ablation matrix shows that price to be dataset-stable at six-to-seven points while all five configurations of the selector cost the same tens of milliseconds. End to end with a production LLM as the reader at the  $4\times$  operating point, Sieve is statistically matched with truncation and the question-agnostic floor on QASPER at  $n=100$  (paired 95% interval  $[-2.1, 10.4]$  on the F1 difference), retaining 61.8% of full-context F1; the relevance-only BM25 reference retains 76.0% with a resolved paired advantage of  $[-4.8]$  F1 points—the same ordering its selector-level recall advantage predicts, now at  $4\times$  the conference sample. The measured GPT-2 perplexity-gate baseline reaches 28.7% evidence recall at  $4\times$  at a scoring cost of 8.2 s per median context on the same container, against Sieve’s tens of milliseconds: a  $320\times$  measured gap at the median operating point that widens to  $255\times$  at a 6,000-word context.

Our contributions are:

- a formulation of long-context compression as budgeted question-conditioned information selection—an explicit selection problem that jointly accounts for relevance coverage, positional reliability, and redundancy under a fixed word budget, with the selector’s own computational cost as a first-class constraint (section 3); to our knowledge, no published selector combines these objectives without hosting a model, an audit we verify entry by entry in Table 1 rather than assert;
- a three-axis evaluation methodology that measures task quality, compression ratio, and the cost of the compressor itself on the same hardware for every method, and a frontier analysis that prices each selection rule on all three axes (section 9);
- the empirical findings the formulation predicts: the margin from truncation to question conditioning is nearly free; relevance-only selection, however implemented (BM25 externally, the relevance term internally), converges to a reference ceiling for evidence recovery; and the positional prior and redundancy penalty are objectives that spend recall rather than recover it—the prior at a dataset-stable six-to-seven paired points, the penalty at no recall cost but also with no measured benefit on the recall metric—a structural trade-off between selection objectives observed from both inside and outside the selector, replicated across datasets and budgets by the full configuration–budget matrix (subsection 8.1);

- Sieve, a no-hosted-model instantiation of the formulation—training-free, model-free, milliseconds of CPU arithmetic, no dependencies beyond NumPy—with a formal analysis of its  $O(W)$  preprocessing,  $O(k n \bar{s})$  budgeted selection, and selection guarantees under a stated lexical-coverage condition;
- a three-benchmark, 1600-record reader-free evaluation with separated paired intervals over six selectors (head, random, stride, TextRank, BM25, and Sieve), a cross-dataset ablation and budget matrix over every configuration at four budgets on all three pools, end-to-end QA accuracy under a production reader at the  $4\times$  operating point, a sample-level correlation study connecting the two protocols, a same-hardware GPT-2 perplexity-gate baseline that instantiates the neural corner’s mechanism and cost floor, an analytical projection onto prefill FLOPs, key-value memory, and dollar cost for LLaMA-class serving, and a robustness study over evidence placement, hyperparameters, and cross-dataset transfer.

Sieve is not a replacement for attention-level efficiency work; it is a low-cost first stage (tens of milliseconds of CPU arithmetic, no hosted model) that composes with KV cache eviction (Zhang et al. 2023; Xiao et al. 2023) and serving optimizations (Dao et al. 2022; Kwon et al. 2023). More broadly, the formulation and the measured frontier outlive the particular selector: any context compressor—statistical or neural—can be priced on the same three axes before justifying its infrastructure.

## 2 Related work

### *Neural token scoring for prompt compression.*

LLMLingua compresses prompts by dropping tokens with high perplexity under a small language model, iterating a coarse-to-fine budget search over the full context (Jiang et al. 2023). LongLLMLingua adds question conditioning, a document-reordering strategy motivated by positional bias, and a token-level budget allocator (Jiang et al. 2024). LLMLingua-2 distills GPT-4 judgments into a 117M-parameter token classifier and frames compression as token classification (Pan et al. 2024). Selective Context filters words, phrases, and sentences by self-information under a language model (Li et al. 2023). All of these compressors invoke at least one forward pass of a neural scorer over the full context, which costs model hosting and GPU time, and none of them is the option a user reaches for when the scorer cannot be hosted.

### *Training-free compressors that still host models.*

A newer family drops the training requirement but keeps the models themselves. Perception Compressor retrieves demonstrations with SentenceBERT similarity against guiding questions and performs semi-guided iterative token compression with LLaMA-2-7B (Tang et al. 2025): it is question-aware and position-aware (a dual-slope ratio allocator distributes compression by position), and it defines the current quality ceiling of the training-free family—at the cost of a hosted 7B scorer plus an embedding model plus pre-generated guiding questions. QASPC is training-free, question-aware, and operates at sentence and clause level with explicit redundancy reduction (Chen et al. 2025); its full text was not accessible at the time of writing, so we compare

against its stated properties rather than a measured replication. A 2025 survey maps the prompt-compression family systematically (Li et al. 2025). In the usage consolidated across this literature, “training-free” means free of training, not free of models. Sieve differs structurally, not component-wise: to our knowledge it is the selector in this family that runs with no model at all, and the question the paper answers is how much of the multi-objective selection problem survives that constraint.

***Learned sentence-level selectors and adaptive budgets.***

Closest to Sieve’s granularity are the learned sentence selectors: RECOMP trains an extractive compressor that selects useful sentences from retrieved documents (plus an abstractive variant) (Xu et al. 2024); CPC trains a context-aware sentence encoder for the same filtering role (Liskavets et al. 2025). AdaComp adaptively sets the compression rate from query complexity and retrieval quality at document granularity (Zhang et al. 2024). These works establish that sentence-level, question-aware selection is a live research problem, and they locate the training cost squarely on the selector. Sieve shares the granularity and the question conditioning but replaces every learned signal with corpus-local statistics, which removes the training set, the training compute, and the served artifact from the selector’s bill of materials.

***Classical selection, ranking, and summarization.***

Maximal marginal relevance formulates extraction as a trade-off between query relevance and redundancy (Carbonell and Goldstein 1998); TextRank ranks sentences by centrality on a similarity graph (Mihalcea and Tarau 2004); TF-IDF and BM25 weighting remain the workhorses of sparse retrieval (Robertson and Zaragoza 2009). Graph machinery also enters recent prompt compression: Prompt-SAW builds relation-aware graphs over the prompt and extracts key information elements for both task-agnostic and task-aware settings (Ali et al. 2024); its scoring is model-light (statistical parsing components rather than a hosted scorer) but it has no positional objective and no budget or cost reporting. Sieve is a question-conditioned instantiation of the classical tradition: IDF-weighted coverage with a positional prior and a word-budget greedy fill. The journal version adds BM25 itself to the baseline set, so that the classical ranking floor is measured rather than asserted: BM25 under the same word budget is a strong question-conditioned, model-free comparator, and the experiments treat it as the reference point that any added selector machinery must justify. Compared with neural compressors, the whole model-free family gives up paraphrase robustness; the measurements below state exactly how much question conditioning recovers and where the lexical signal runs out.

***Positional bias.***

Liu et al. (2023) showed that answer accuracy in long contexts follows a U shape in the position of the evidence. The observation motivates recency-aware cache designs such as StreamingLLM (Xiao et al. 2023) and the reordering and ratio-allocation mechanisms of LongLLMLingua and Perception Compressor; we use it to place a prior on sentence positions inside a statistical selector, and we measure the selector-level analogue of the same curve.

**Table 1** Selector design space along seven dimensions: question conditioning (Q), positional awareness (P), explicit redundancy control (R), model-free at inference (MF), training-free (TF), sentence-level intact units (SL), and explicit budget formulation with measured compressor cost (BF). ✓ = yes; × = no; \* = partial or qualified (mechanism present but limited, or model-light machinery rather than a hosted scorer); ? = not verifiable from the sources accessible at the time of writing (QASPC’s properties as stated by its abstract). Every entry is auditable against the cited source. To our knowledge, Sieve is the only selector that is simultaneously question-conditioned, position-aware, redundancy-controlling, and entirely model-free with its own cost measured on the frontier.

| Selector                                 | Q | P | R | MF | TF | SL | BF |
|------------------------------------------|---|---|---|----|----|----|----|
| MMR (Carbonell and Goldstein 1998)       | ✓ | × | ✓ | ✓  | ✓  | ✓  | ×  |
| BM25 (Robertson and Zaragoza 2009)       | ✓ | × | × | ✓  | ✓  | ✓  | *  |
| TextRank (Mihalcea and Tarau 2004)       | × | × | * | ✓  | ✓  | ✓  | ×  |
| Prompt-SAW (Ali et al. 2024)             | ✓ | × | * | *  | ✓  | ✓  | ×  |
| Selective Context (Li et al. 2023)       | × | × | * | ×  | ✓  | *  | ×  |
| LLMLingua (Jiang et al. 2023)            | × | × | × | ×  | ✓  | ×  | ✓  |
| LongLLMLingua (Jiang et al. 2024)        | ✓ | ✓ | × | ×  | ✓  | ×  | ✓  |
| LLMLingua-2 (Pan et al. 2024)            | × | × | × | ×  | ×  | ×  | ✓  |
| RECOMP (Xu et al. 2024)                  | ✓ | × | × | ×  | ×  | *  | ×  |
| CPC (Liskavets et al. 2025)              | ✓ | × | * | ×  | ×  | ✓  | ×  |
| AdaComp (Zhang et al. 2024)              | ✓ | × | × | *  | ✓  | ×  | ✓  |
| Perception Compressor (Tang et al. 2025) | ✓ | ✓ | × | ×  | ✓  | ×  | *  |
| QASPC (Chen et al. 2025)                 | ✓ | ? | ✓ | ?  | ✓  | ✓  | ?  |
| Sieve (this work)                        | ✓ | ✓ | ✓ | ✓  | ✓  | ✓  | ✓  |

### What is new here.

Table 1 locates the prior selectors on seven dimensions across thirteen entries, and the delta it exposes is threefold, stated with the audit’s scope conditions. First, positional-aware and redundancy-aware selection exist only in selectors that host at least one model (LongLLMLingua, Perception Compressor) or that train one (RECOMP, CPC); QASPC states both properties but its model-freeness is not verifiable from accessible sources, and Prompt-SAW’s graph machinery is model-light without either axis; to our knowledge, no verified published selector combines them with model-free operation, and that combination is what lets Sieve run in milliseconds on a CPU core. Second, no prior work treats the compressor’s own cost as a measured axis alongside quality and compression ratio; the budget mechanisms in the LLMLingua family and AdaComp control *how much* is kept, not *what the keeping costs*, and the same-hardware gate measurements of section 9 are, to our knowledge, the first to price that axis for both a statistical and a neural selector. Third, the formulation of selection as a joint three-objective problem under a budget (relevance, positional reliability, redundancy) is absent from prior work even where the corresponding mechanisms appear as isolated heuristics; section 3 states it, and the ablations and the BM25 cross-check of section 8 measure its internal trade-offs. The table is also the paper’s map for future work: any new compressor, statistical or neural, can be placed on it and on the measured frontier.



### 3 Problem formulation

#### 3.1 Setting

A query instance is a pair  $(C, Q)$  where  $C = (s_1, \dots, s_n)$  is a context of  $n$  sentences with  $W = \sum_i |s_i|$  words and  $Q$  is a question. A selector-compressor  $\kappa$  maps  $(C, Q)$  to a subsequence  $C' = \kappa(C, Q) \subseteq C$  with  $\|C'\| \leq B$  words, where  $B = W/r$  for a target compression ratio  $r \geq 1$ . A reader LLM  $R$  receives  $C'$  and  $Q$  and produces an answer  $a = R(C', Q)$ . Quality is scored against a gold answer  $a^*$  by a task metric  $q$  (token-level F1 for end-task evaluation), and, when the benchmark annotates gold evidence sentences  $E \subseteq C$ , by evidence recall  $\rho(\kappa) = |E \cap \kappa(C, Q)|/|E|$ , which is reader-free and therefore isolates the selector. The two quality measures are related but not interchangeable; [section 11](#) measures how loosely the first bounds the second.

#### 3.2 The three-way trade-off

Comparing compressors requires fixing three axes, not one:

- (i) *Quality*  $q$ : end-task agreement with  $a^*$ , or evidence recall  $\rho$  when the question is which sentences survive.
- (ii) *Compression*  $r$ : the realized context reduction  $W/\|C'\|$ , which converts directly into prefill FLOPs, cache memory, and per-token fees.
- (iii) *Compressor cost*  $c(\kappa)$ : the resources consumed by  $\kappa$  itself: hosted model parameters, accelerator requirements, and wall-clock latency per context.

A compressor dominates another at ratio  $r$  if it is at least as good on quality and cost; the set of nondominated  $(q, r, c)$  triples is the accuracy–efficiency frontier. The formulation separates families that a single-axis view conflates: truncation occupies the  $(q, r, c)$  corner with  $c \approx 0$  but question-agnostic quality; neural gates occupy the high- $q$  corner with  $c$  bounded below by one forward pass of a hosted scorer over the full context; Sieve is designed for the corner where  $q$  must be bought without any hosted model, and BM25 sits in the same corner with the relevance term alone. Every table in [section 7–section 10](#) reports all three axes for every method, and [section 9](#) plots the frontier explicitly.

#### 3.3 Cost accounting for selectors

For a selector with parameters  $\theta$  we distinguish three cost terms:  $c_{\text{host}}$  (whether a model must be hosted at all),  $c_{\text{flop}}$  (arithmetic work in FLOPs or equivalent elementary operations), and  $c_{\text{lat}}$  (measured wall-clock per context). For neural gates built on dense transformer scorers (the perplexity-gate family) the floor is one forward pass of the scorer  $M_s$  over the full context,  $c_{\text{flop}} \geq 2|M_s|L$  for  $L$  context tokens, with  $c_{\text{host}} \neq 0$ ; coarse-to-fine budget searches repeat the pass. For Sieve, all three terms are the object of [section 5](#): the selector is a fixed small set of arithmetic passes over the token stream with  $c_{\text{host}} = 0$ ,  $c_{\text{flop}} = O(W)$  elementary operations for preprocessing and  $O(k n \bar{s})$  for the greedy fill, and  $c_{\text{lat}}$  measured in the tens of milliseconds on one



CPU core. Latency and FLOPs are reported for every method on the same hardware in [section 9](#), so the frontier coordinates are measured, not asserted.

### 3.4 Evaluation protocols

We use three protocols, each isolating one part of the pipeline. *End-task protocol*: the reader answers from  $C'$ ; we report EM and F1 with percentile bootstrap intervals over examples, and paired bootstrap intervals for method differences. *Selector protocol*: on benchmarks with annotated evidence we report  $\rho$  at matched word budgets, with bootstrap intervals and paired intervals for the Sieve-versus-baseline differences; the same protocol runs every ablation configuration of [section 8](#) at every budget on every pool, which is what produces the configuration–budget–dataset matrix of [subsection 8.1](#); this protocol involves no reader and scales to thousands of records, which is what gives the multi-dataset comparison its power. *Cost protocol*: wall-clock of  $\kappa$  alone, measured on fixed hardware with the reader excluded, plus the realized compression ratio; for hosted scorers we additionally report the analytic GPU floor  $2|M_s|L/(\text{peak} \cdot \text{MFU})$ . The three protocols are reported separately and never mixed: an end-task number is never compared against a selector-level number without saying so.

## 4 The Sieve framework

### 4.1 Overview

Sieve treats compression as budgeted sentence selection. Given  $(C, Q)$  and a ratio  $r$ , it (i) segments the context into sentences, (ii) computes IDF statistics over the context itself, (iii) scores each sentence by a convex combination of question relevance and a U-shaped positional prior, (iv) fills the word budget  $B = W/r$  greedily under a redundancy penalty, and (v) emits the selected sentences in their original order. No step consults any model, corpus, or learned parameter; the statistics are computed from the instance alone. [Figure 1](#) summarizes the pipeline.

### 4.2 Scoring

Each sentence  $s_i$  receives a base score

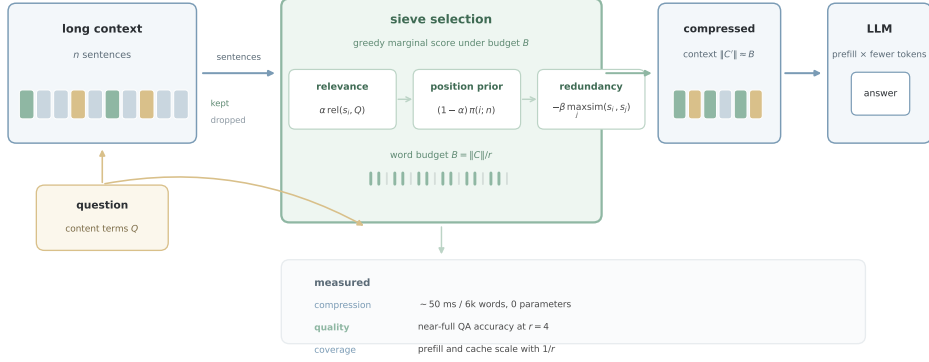
$$\text{base}_i = \alpha \text{rel}_i + (1 - \alpha) \pi_i, \quad (1)$$

a convex combination of query relevance and a positional prior.

#### *Question relevance.*

Tokenize  $Q$  into content terms (lowercased, stopword-filtered). Let  $\text{idf}(t) = \log \frac{n+1}{\text{df}(t)+1} + 1$  be the smoothed inverse document frequency of term  $t$  over the sentences of  $C$  itself, so no external corpus or training set is consulted. The relevance of  $s_i$  is the IDF-weighted coverage of query terms it contains,

$$\text{rel}_i = \frac{1}{\sqrt{|s_i|}} \sum_{t \in Q \cap s_i} \text{tf}(t, Q) \text{idf}(t), \quad (2)$$



**Fig. 1** The Sieve pipeline. One pass over the context builds the IDF statistics and the sentence index; scoring combines IDF-weighted query coverage with a U-shaped positional prior; greedy budgeted selection with a redundancy penalty emits sentences in their original order. No model, tokenizer, or external corpus is involved.

normalized by the square root of sentence length to suppress a bias toward long sentences. Query terms are counted with saturation  $\text{tf}(\cdot, Q) \in \{0, 1, 2\}$  capped at two.

#### *Positional prior.*

Let  $x_i = (i - 1)/(n - 1) \in [0, 1]$  be the relative position of  $s_i$ . The prior is U-shaped,

$$\pi_i = \text{floor} + (1 - \text{floor}) |2x_i - 1|^\gamma, \quad (3)$$

so the first and last sentences receive weight 1 and the middle receives floor. The shape encodes the primacy and recency advantage that placement studies of long-context models report (Liu et al. 2023);  $\gamma$  controls how sharply the middle is penalized. With floor = 1 the prior is flat and ablates away.

### 4.3 Budgeted greedy selection

The selector fills the word budget greedily. At each step it picks the sentence maximizing marginal score

$$\text{score}(i | S) = \text{base}_i - \beta \max_{j \in S} \text{sim}(s_i, s_j), \quad (4)$$

where  $S$  is the already-selected set and  $\text{sim}(\cdot, \cdot) \in [0, 1]$  is IDF-weighted cosine similarity between sentence token vectors. The penalty of weight  $\beta$  skips near-duplicate sentences, which are common in the methods and related-work sections of scientific text. Selection stops when the budget is exhausted; the output concatenates selected sentences in original order. Keeping order matters: the reader sees a coherent excerpt, and the positional prior that shaped selection also shapes what the reader finally sees. [algorithm 1](#) gives the complete procedure.

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**Algorithm 1:** Sieve selection

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**Input:** context sentences  $s_1, \dots, s_n$ , question  $Q$ , budget  $B$ , weights  $\alpha, \beta, \text{floor}, \gamma$   
split  $C$  into sentences; tokenize; build IDF statistics over sentences;  
 $\text{rel}_i \leftarrow (2)$  and  $\pi_i \leftarrow (3)$  for each  $i$ ;  
 $S \leftarrow \emptyset$ ;  $\text{used} \leftarrow 0$ ;  
**while**  $\text{used} < B$  **do**  
     $i^* \leftarrow \arg \max_{i \notin S} \text{base}_i - \beta \max_{j \in S} \text{sim}(s_i, s_j)$  ;      // greedy MMR step  
    **if**  $\text{used} + |s_{i^*}| > B$  **then**  
        **break** ;      // budget exhausted  
    **end**  
     $S \leftarrow S \cup \{i^*\}$ ;  $\text{used} \leftarrow \text{used} + |s_{i^*}|$ ;  
**end**  
**return** the sentences of  $S$  in original index order

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## 4.4 Configuration

Four hyperparameters govern Sieve:  $\alpha$  trades relevance against position,  $\beta$  sets the redundancy penalty,  $\text{floor}$  fixes the middle weight, and  $\gamma$  shapes the prior. We grid-searched  $\alpha \in \{0.3, 0.6, 0.9\}$  on 20 development examples disjoint from the test split and fixed  $\beta = 0.3$ ,  $\text{floor} = 0.35$ ,  $\gamma = 1.5$  without tuning. The selected  $\alpha = 0.3$  leans on the positional prior, a choice the placement analysis of [section 10](#) is consistent with. Sensitivity to all four values is reported in [section 10](#), and (central to the generalization claim of [section 12](#)) the same configuration is applied unchanged to all three datasets: no per-dataset tuning is performed anywhere in this paper.

## 5 Theoretical analysis

This section states what the design guarantees and what it deliberately does not. Proofs are short and appear in [Appendix B](#); the interest is in making the assumptions explicit, because the empirical sections then measure how far real benchmarks satisfy them.

### 5.1 Cost: one arithmetic pass, no hosted model

**Proposition 1** (Cost of Sieve). *Let  $W$  be the context word count,  $n$  the number of sentences,  $k$  the number selected, and  $\bar{s}$  the mean sentence length in tokens. Segmentation, tokenization, and IDF construction take  $O(W)$  elementary word-level operations; relevance and the prior take  $O(W + n|Q|)$ ; the greedy loop takes  $O(k n \bar{s})$  in the worst case, each step dominated by sparse-counter similarity evaluations. Peak memory is the token counters of the input. No parameter, model, or external corpus is consulted:  $c_{\text{host}} = 0$ .*

The proof is immediate from the algorithm: each preprocessing stage touches each token a constant number of times, and the greedy loop performs at most  $k$  passes

over  $n$  candidates whose similarity evaluation costs  $O(\bar{s})$  on sparse counters. At the operating point of our experiments ( $W \approx 6,000$ ,  $n \approx 300$ ,  $k \approx 75$ ) the bound is a few million elementary operations, against a measured 53.7ms on one CPU core of the evaluation container (section 9). The greedy stage, not the preprocessing, dominates at this operating point, which is why we state the two terms separately rather than call the whole selector linear.

**Proposition 2** (Floor of scorer-based compression). *Any compressor that scores the context with a dense decoder-only transformer  $M_s$  of  $|M_s|$  parameters over  $L$  context tokens, via a full forward pass—the mechanism of the perplexity-gate family—performs at least  $2|M_s|L$  FLOPs per compression call, before any budget search that repeats the pass, and incurs  $c_{\text{host}} \neq 0$ . Scorers of other architectures (sparse mixers, linear attention) have analogous architecture-specific floors that this proposition does not quantify.*

This is the standard forward-pass FLOP count (Kaplan et al. 2020), stated for the dense decoder-only transformer that the perplexity-gate family and our gate baseline instantiate; architectures with cheaper scoring exist, and the proposition’s scope is the family actually measured in section 9. For GPT-2-small ( $|M_s| = 124\text{M}$ ) at  $L \approx 8,000$  tokens, the floor is  $1.98 \times 10^{12}$  FLOPs: at least five orders of magnitude above Sieve’s arithmetic bound at the same context, and the floor does not include hosting, tokenization matched to the scorer, or the coarse-to-fine repetitions that budget searches add. section 9 measures both sides on identical hardware rather than leaving the comparison at the FLOP level.

## 5.2 Selection: what question conditioning guarantees

We now state the condition under which relevance scoring provably recovers evidence, and contrast it with truncation. The condition is about the instance, not the method.

**Condition 1** (Lexical coverage). *Instance  $(C, Q, E)$  satisfies lexical coverage if every gold evidence sentence weakly dominates every non-evidence sentence in IDF-weighted query coverage:  $\text{rel}(e) \geq \text{rel}(s)$  for all  $e \in E$ ,  $s \in C \setminus E$ .*

**Proposition 3** (Recall of relevance-only selection). *Under 1, if the budget satisfies  $B \geq \sum_{e \in E} |e|$ , then relevance-only selection ( $\alpha = 1$ , flat prior,  $\beta = 0$ ) attains evidence recall  $\rho = 1$ : every evidence sentence is selected before any non-evidence sentence it dominates, and all fit within  $B$ .*

**Proposition 4** (Recall of head truncation). *Let the position of each gold evidence sentence be uniform on  $\{1, \dots, n\}$  and independent across sentences, and let truncation keep the prefix of  $\lceil n/r \rceil$  sentences. Then the expected evidence recall is  $\mathbb{E}[\rho_{\text{head}}] = \min(1, 1/r)$ .*

Two readings follow. First, the separation between question-conditioned and question-agnostic selection grows linearly in  $r$ : under 1 with sufficient budget the gap

is  $1 - 1/r$ , which is 50 points at  $r = 2$ , 75 at  $r = 4$ , and 94 at  $r = 16$ . This is exactly the shape the ratio sweeps of [section 7](#) measure on all three datasets. Second, the condition is strong, and real data satisfies it only partially: paraphrased evidence, distractors that share query terms, and multi-hop questions whose terms spread across sentences all violate it. The ablations of [section 8](#) quantify the departure: relevance-only recall on QASPER is well below the  $\rho = 1$  of the idealized condition; and [section 12](#) measures how the departure varies by dataset. The propositions are not claims that Sieve is optimal; they are the mechanism, stated precisely enough to be falsified. They also delimit what BM25 can claim: BM25 is a relevance-only selector up to term saturation and length normalization, so [3](#) is the guarantee it approximates, and [1](#) does not apply to it (BM25 is likewise model-free), which is why the empirical sections measure BM25 rather than bound it.

### 5.3 The positional prior as a reliability allocation

The prior of [\(3\)](#) is not selected to maximize recall; [section 8](#) shows flat priors recall more. It allocates the selection’s reliability budget to the positions where readers attend most reliably. If reader accuracy in using a retained sentence is a U-shaped function  $u(x)$  of its position  $x$ , as placement studies report ([Liu et al. 2023](#)), then a selector that must discard sentences maximizes expected usable evidence by discarding from the middle first, which is what the prior does: with  $\text{floor} < 1$ , middle sentences enter the budget only after end-position sentences of comparable base score. The cost is explicit and measured in [section 10](#): when the only evidence copy sits mid-document, the prior discards it. The design treats selector recall and reader reliability as exchangeable under a stated profile rather than optimizing either alone, and the  $\text{floor}$  hyperparameter exposes the exchange rate to the deployer. BM25, which lacks the prior, occupies the opposite end of this exchange: maximal raw recall, no placement control. Which end a deployment should choose is an empirical question the end-task protocol addresses.

### 5.4 What the design does not guarantee

Three limits are worth stating before the measurements, because the measurements confirm all three. *Lexical reach*: [1](#) fails outright for paraphrased or multilingual evidence, and IDF weights numbers and code identifiers crudely. *Budget tightness*: when  $B < \sum_{e \in E} |e|$  no selector attains  $\rho = 1$ , and the ratio sweep measures how each degrades. *Reader variance*: end-task quality couples the selector to the reader’s own positional profile, which the selector models only through the U-shape; our re-measurement of the reader-side profile ([subsection 10.1](#)) found it flat at the median  $\sim 1,000$ -word scale of the synthetic protocol, so the prior’s reader-side justification rests on the cited longer-context evidence ([Liu et al. 2023](#)), and the exchange rate it assumes is priced empirically rather than proven—a scope limitation [section 13](#) returns to.

## 6 Experimental setup

### 6.1 Benchmarks and pools

We use three benchmarks, chosen because each annotates the gold evidence, which the reader-free protocol of [subsection 3.4](#) requires, and because together they span abstract and entity-named question styles. Long-context suites without evidence annotations (LongBench ([Bai et al. 2023](#)), MuSiQue ([Trivedi et al. 2022](#)) in its distractor form) cannot support the recall protocol and are therefore out of scope for the selector-level claims.

#### ***QASPER.***

QASPER ([Dasigi et al. 2021](#)) contains information-seeking questions over full NLP papers. From the train split’s paper-level records we select questions whose gold answers are extractive spans, short free-form strings, or yes/no judgments, cap each context at 6,000 words (the cap binds on papers that exceed it), and keep records whose annotated evidence sentences are locatable in the segmented context. The resulting recall pool contains 1000 records (median 3538 words, 90th percentile 5,576), from which the end-task test split (100 records: the pool’s fixed first-hundred prefix, the same records the GPT-2 gate scored, so per-record recall and end-task F1 join exactly for the correlation study of [section 11](#)) and the 20-record development split (ten further QASPER records plus ten HotpotQA distractor records) are drawn disjointly and fixed before any test evaluation; the development split mixes the two corpora for tuning diversity, while the test split is QASPER only. The official QASPER test labels are not public, which is why all held-out sampling comes from the train split.

#### ***HotpotQA.***

HotpotQA ([Yang et al. 2018](#)) in the distractor setting pairs each question with ten paragraphs, two of which contain the supporting facts. We flatten the paragraphs into a single context under the same 6,000-word cap and keep records whose supporting-fact sentences are locatable. The recall pool contains 300 records (median 888 words).

#### ***2WikiMultihopQA.***

2WikiMultihopQA ([Ho et al. 2020](#)) builds compositional comparison and inference questions over Wikipedia with annotated reasoning chains. The same flattening and locatability protocol yields a pool of 300 records (median 602 words). Its dev split provides the examples; no test labels are used.

The three pools total 1600 records. All are processed by the same segmenter, the same locatability rule (evidence sentence prefix-matches a context sentence), and the same word-budget convention; no per-dataset tuning of any kind is performed.

### 6.2 Reader and decoding

All end-task conditions use the same reader: GLM-4-Plus served through its public API ([GLM Team et al. 2024](#)), temperature 0, maximum 96 answer tokens, and a fixed instruction to answer with the shortest span from the context or yes/no. The reader

sees only the compressed (or full) context and the question. We record the served token counts of every call. End-task evaluation is scoped to the QASPER test split at the  $4\times$  operating point; the selector-level protocols carry the multi-dataset claims.

### 6.3 Baselines

All selectors share Sieve’s word-budget convention so that the comparison isolates the selection rule. Full context uses  $C$  unchanged. Head truncation keeps the first  $B$  words. Random selects sentences uniformly under the budget with two seeds. Stride keeps every  $k$ -th sentence with  $k$  chosen to meet the budget. TextRank (Mihalcea and Tarau 2004) ranks sentences by PageRank on an IDF-weighted cosine graph and packs the top-ranked sentences under the budget; it is a strong model-free summarizer baseline that does not see the question, and the strongest question-agnostic baseline we evaluate.

#### *BM25 sentence ranking.*

BM25 (Robertson and Zaragoza 2009) is the classical question-conditioned baseline: the question is the query, each sentence is a document, IDF is smoothed over the sentences of the context itself (matching Sieve’s corpus-local statistics), and the top-ranked sentences are packed under the same budget and emitted in original order. It is the question-aware model-free reference point of the study, the bar that any selector machinery beyond relevance ranking (positional priors, redundancy penalties, greedy marginal search) must justify. We run it on all three pools at every ratio.

#### *GPT-2 perplexity gate (neural baseline).*

To compare against the neural compressors on measured terms, we implement the perplexity gate of LLMingua-style compression (Jiang et al. 2023) at sentence granularity, the same output granularity as Sieve, so the comparison isolates the scorer and not the unit. The scorer is GPT-2 small (124M parameters) (Radford et al. 2019), dynamically quantized to int8. For each record, one forward pass scores every sentence by the mean negative log-likelihood of its tokens, with the question prepended to the first window so the gate is question-conditioned in the LongLLMingua sense (Jiang et al. 2024); sentences are packed into at most 1,024-token windows of whole sentences, which is GPT-2’s position limit. Selection keeps the lowest-perplexity sentences under the same budget and preserves order. The pass is the floor of what any scorer-based compressor pays; coarse-to-fine budget searches repeat it. We run the gate on 100 QASPER records (the first 100 of the recall pool, a fixed prefix) and measure its scoring latency on the same container as every other timing number in this paper. We do not run LLMingua-2’s distilled token classifier: its checkpoint requires hosting a different architecture, and the GPT-2 gate already represents the mechanism and the cost floor of that family; section 13 returns to what that comparison cannot claim.

### 6.4 Metrics and statistics

We report exact match (EM) and token-level F1 with SQuAD-style normalization (Yang et al. 2018) against gold answers, plus the percentage of full-context F1 retained



by each compressor. For the selector protocol we report evidence recall at matched budgets. Uncertainty is a percentile bootstrap over records with 1,000 resamples, reported as 95% intervals on the means, together with paired bootstrap intervals for the Sieve-versus-baseline differences per dataset; the random baseline additionally reports the standard deviation across two selection seeds. Timing numbers are medians of five runs after one warm-up (three for the gate). The recall and end-task protocols are never pooled: where both appear, their sample sizes are stated separately.

## 6.5 Implementation and compute

The selector is a single Python module with NumPy as its only dependency beyond the standard library. All journal-version experiments ran in a container with 2 vCPUs and 4 GB of memory; the GPT-2 gate ran in the same container on the same cores (int8, two threads), so every latency comparison in [section 9](#) is same-hardware. No GPU was used for any measurement reported in this paper; the A100 floor for the gate is analytic, computed from  $2|M_s|L$  FLOPs at 40% utilization, and the projection of [subsection 9.5](#) is likewise analytical with stated assumptions. The repository regenerates every table and figure from raw checkpoints with one command per artifact.

### *Evaluation questions.*

The experiments are organized around six questions, each carried by a named section: **EQ1**—does question conditioning beat question-agnostic selection at matched budgets? ([subsection 7.2](#), [subsection 7.3](#)); **EQ2**—what does each selection objective cost and buy, and does the internal relevance term agree with an external relevance ranker? ([section 8](#), [subsection 8.1](#)); **EQ3**—what does the trade cost on the compressor’s own compute axis, and where does a neural scorer’s cost floor sit? ([section 9](#)); **EQ4**—is the behavior stable across evidence placement, hyperparameters, and datasets without retuning? ([section 10](#)); **EQ5**—does the selector-level ordering transfer to end-task answers under a production reader? ([subsection 7.1](#)); **EQ6**—how tightly does the selector-level metric predict the end-task one? ([section 11](#)). The claim hierarchy is fixed accordingly: multi-dataset claims are selector-level (EQ1–EQ4, 1600 records); reader-level claims are single-reader, single-benchmark, and single-operating-point (EQ5–EQ6,  $n=100$ ); and the serving projection of [subsection 9.5](#) is analytical and labels itself as such. No conclusion below mixes layers: an end-task number is never compared against a selector-level number without saying so, and a projection is never cited as a measurement.

## 7 Main results

### 7.1 End-task question answering at $4\times$

[Table 2](#) reports the end-task comparison at  $r = 4$  on the test split. Three observations. First, every compressor loses accuracy relative to the full context, and the loss is large: the full-context interval is wide at  $n=100$ , but its point estimate (33.6 F1) sits well above every compressed condition, which is the expected price of removing three quarters of the context. Second, Sieve and head truncation cannot be separated

**Table 2** Question answering at a  $4\times$  compression on the QASPER test split ( $n=100$  per method after exclusions, see [subsection 6.1](#)), reader GLM-4-Plus at temperature 0. EM and token-level F1 in percent; brackets are bootstrap 95% intervals;  $\Delta F1$  is the gap to full context. Prompt tokens are the served means. Sieve, truncation, and random selection overlap within paired intervals; stride and TextRank sit below; BM25 retains the most F1 among the compressors. Paired intervals are reported per baseline in the text; orderings among the question-agnostic compressors remain underpowered at  $n=100$ , while the BM25 gap is resolved. The GPT-2 gate was not run under this protocol; it is compared under the selector protocol ([subsection 7.2](#), [section 9](#)). BM25 is included under this protocol.

| Method       | EM $\uparrow$ | F1 $\uparrow$ | $\Delta F1$ | Retain | prompt tokens |
|--------------|---------------|---------------|-------------|--------|---------------|
| Full context | 11.0          | <b>33.6</b>   | –           | 100%   | 5181          |
| Head         | 4.0           | <b>16.7</b>   | 16.8        | 49.9%  | 1334          |
| Random       | 5.0           | 20.5          | 13.1        | 61.0%  | 1354          |
| Stride       | 3.0           | 14.1          | 19.5        | 42.0%  | 1201          |
| TextRank     | 5.0           | 18.4          | 15.2        | 54.7%  | 1392          |
| BM25         | 4.0           | 25.5          | 8.0         | 76.0%  | 1320          |
| Sieve        | 3.0           | 20.7          | 12.8        | 61.8%  | 1315          |

at  $n=100$  in either direction: the paired bootstrap interval for the F1 difference is  $[-2.1, 10.4]$ , with point estimate 4.0 points and half-width 6.3 points. On questions about a paper’s overall setup, the evidence often sits in the opening sections, which is the regime where a fixed prefix is already competent. Third, stride and TextRank sit below Sieve in point estimate (paired difference of 2.4 points against TextRank), while random selection is level with it (0.3 points)—the question-agnostic floor clears truncation but does not separate from the full selector. These paired intervals include zero at  $n=100$  (half-widths 4.9 and 6.1 points); we report the point estimates with their widths rather than claim separations the design cannot resolve. Against the question-conditioned lexical baseline the paired interval is  $[-8.6, -1.4]$  (point estimate -4.8 points, half-width 3.6): the interval excludes zero in BM25’s favor, so the trade-off narrative is consistent at both layers—raw lexical relevance dominates both the recall axis and the end-task axis at this operating point, with BM25 retaining 76.0% of full-context F1 against 61.8% for Sieve. What Sieve still buys at the same token budget is placement-shaped evidence and redundancy control; their recall price is quantified by the ablation ([section 8](#)) and reappears here as the end-task gap to the relevance-only reference. The end-task picture is therefore that Sieve is statistically matched with truncation and the question-agnostic floor at  $n=100$ , while the relevance-only BM25 reference holds a resolved paired advantage—the same ordering the selector-level recall protocol reports at three datasets’ scale. Where the question actually matters, the picture sharpens at the scale the selector protocol can afford, which is the subject of the next two subsections.

**Table 3** Gold evidence sentence recall at a  $4\times$  budget under the reader-free protocol (no LLM calls; brackets are bootstrap 95% intervals over records). Sieve uses the same configuration on all three datasets.  $\Delta$  rows are paired 95% intervals for the difference in recall points. BM25 is the question-conditioned, model-free reference; the question-agnostic selectors cluster far below.

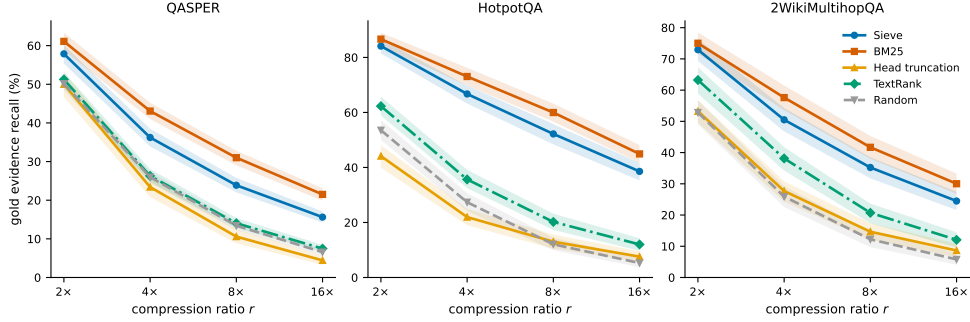
|                       | QASPER ( $n=1000$ )      | HotpotQA ( $n=300$ )     | 2Wiki ( $n=300$ )        |
|-----------------------|--------------------------|--------------------------|--------------------------|
| Sieve                 | <b>36.2</b> [33.9, 38.5] | <b>66.7</b> [63.2, 69.8] | <b>50.5</b> [46.8, 54.3] |
| BM25                  | 43.1 [41.0, 45.2]        | 73.1 [69.8, 76.2]        | 57.6 [53.6, 61.5]        |
| Head                  | 23.4 [20.8, 26.0]        | 22.0 [19.0, 25.0]        | 27.7 [24.3, 30.8]        |
| Random                | 26.0 [24.5, 27.4]        | 27.4 [24.1, 30.7]        | 25.9 [22.6, 29.2]        |
| Stride                | 22.2 [21.2, 23.2]        | 26.6 [23.4, 29.7]        | 21.4 [18.4, 24.4]        |
| TextRank              | 26.3 [24.7, 28.1]        | 35.6 [32.5, 39.1]        | 38.1 [34.2, 41.8]        |
| $\Delta$ (Sieve–Head) | [10.0, 15.3]             | [40.7, 48.9]             | [18.2, 27.2]             |
| $\Delta$ (Sieve–BM25) | [-8.7, -5.0]             | [-9.0, -3.7]             | [-10.5, -4.0]            |

## 7.2 Evidence retention at scale: three datasets

Table 3 is the paper’s central selector-level result, and the conference version’s single-dataset finding survives the scaling intact and sharpened. On QASPER at  $n=1000$ , Sieve retains 36.2% of the annotated evidence against 23.4% for head truncation, and the intervals, which at the conference version’s  $n = 191$  only touched at their edges, are now separated, with the paired difference at 12.8 points [10.0, 15.3]. On HotpotQA the margin more than triples: 66.7% against 22.0%, a paired difference of 44.8 points [40.7, 48.9]. On 2WikiMultihopQA the margin is 22.8 points [18.2, 27.2]. Every question-agnostic selector sits below Sieve on every dataset, and every paired interval excludes zero by a wide margin.

The BM25 rows give the table its second axis, and the finding is reported as measured: BM25 retains *more* raw evidence than the full Sieve configuration on all three datasets, by  $|-6.8|$  paired points on QASPER [-8.7, -5.0], -6.3 on HotpotQA, and -7.1 on 2WikiMultihopQA. The result is not an anomaly but the direct empirical counterpart of 3: BM25 is relevance ranking distilled, and relevance ranking is exactly the term of Sieve that recovers evidence. The selector’s own ablations (section 8), run on all three pools, say the same thing from the inside on every one: stripping the selector to relevance alone reaches 43.6%, 71.9%, and 56.8%, within a point of BM25’s 43.1%, 73.1%, and 57.6% on the respective pools, and flattening the positional prior recovers six-to-seven paired points on every dataset. What BM25 and the relevance-only ablation give up is the subject of the placement analysis (section 10): retained evidence concentrated where readers attend least reliably, and no redundancy control under tight budgets. The end-task protocol, not the recall protocol, is where that trade is decided, and section 11 measures how loosely the two protocols are connected.

Two regularities in the table deserve emphasis. First, head truncation lands at 22–31% recall on all three datasets, close to the  $1/r = 25\%$  that 4 predicts for uniformly placed evidence, and section 12 confirms that evidence positions are close to uniform



**Fig. 2** Gold evidence sentence recall as a function of the compression ratio  $r$  at matched word budgets, one panel per dataset (shaded regions are bootstrap 95% intervals). The pattern is the same on all three: truncation is competitive at  $r = 2$  and collapses as the budget tightens; the question-conditioned selectors (Sieve, BM25) degrade far more slowly; and the gap over every question-agnostic selector is separated by the intervals from  $r = 4$  onward.

on all three pools. The prediction is not a strawman: it is what a fixed prefix mechanically achieves, and the measured agreement is a useful validation that the recall protocol measures what it claims. Second, Sieve’s margin over the question-agnostic selectors differs by dataset, and the ordering tracks the lexical signal in the question. HotpotQA and 2WikiMultihopQA questions name the entities whose sentences carry the answer, so IDF-weighted coverage fires strongly; QASPER’s questions are more often abstract (what, how, why), so the signal is weaker and the margin smaller. The same Sieve configuration is used throughout; the datasets, not the method, vary.

### 7.3 Evidence retention under increasing compression

Figure 2 sweeps the budget from  $r = 2$  to  $r = 16$  under the evidence-recall protocol on all three pools. The qualitative pattern from the conference version’s single dataset replicates on the two additions, and BM25 tracks it from above. At  $r = 2$ , every method retains most of the evidence and the compressors are within noise of one another (57.9 for Sieve against 50.0 for truncation and 61.1 for BM25 on QASPER; 84.1 against 44.1 on HotpotQA). The separation grows as the budget tightens. On QASPER at  $r = 8$ , Sieve retains 23.9% of the annotated evidence against 10.6% for truncation and 14.0 for TextRank, with BM25 at 31.0; on HotpotQA the same ratio is 52.2 against 13.0; on 2WikiMultihopQA 35.2 against 14.7. At  $r = 16$  the gaps reach 15.6 against 4.4 on QASPER and 38.6 against 7.6 on HotpotQA, with non-overlapping intervals throughout. The mechanism is the design intent, now confirmed across datasets: relevance scoring keeps pulling question-covering sentences out of the whole document while a fixed prefix shrinks into the first quarter of the text, and the same mechanism that is interchangeable at generous budgets dominates at tight ones, where compression actually matters. This is the empirical shape of 3 and 4: the gap grows in  $r$  as the propositions predict, though the absolute level of every method’s recall stays below the idealized  $\rho = 1$  of 1, by an amount that measures how far real benchmarks satisfy lexical coverage.

**Table 4** Selector-level ablations at  $r = 4$  on all three recall pools (bootstrap 95% intervals; the same protocol as Table 3). Question conditioning carries the most recall everywhere and its paired margin scales with how entity-named the questions are; the positional prior costs a stable six-to-seven paired points on every dataset, the price paid for placement; the redundancy penalty is recall-neutral on all three; and the internal relevance-only configuration agrees with external BM25 within a point on every dataset.

|                                                 | QASPER                   | HotpotQA                 | 2WikiMultihopQA          |
|-------------------------------------------------|--------------------------|--------------------------|--------------------------|
| Sieve (full)                                    | <b>36.2</b> [33.9, 38.5] | <b>66.7</b> [63.2, 69.8] | <b>50.5</b> [46.8, 54.3] |
| BM25 (external reference)                       | 43.1 [41.0, 45.2]        | 73.1 [69.8, 76.2]        | 57.6 [53.6, 61.5]        |
| no question conditioning ( $\alpha=0$ )         | 19.9 [17.5, 22.3]        | 28.2 [24.9, 31.6]        | 27.4 [23.9, 30.8]        |
| flat positional prior (floor=1)                 | 43.3 [41.4, 45.1]        | 72.8 [69.5, 75.6]        | 56.9 [53.0, 60.9]        |
| no redundancy penalty ( $\beta=0$ )             | 36.3 [34.0, 38.5]        | 67.1 [63.6, 70.1]        | 51.0 [47.3, 54.5]        |
| relevance only ( $\alpha=1$ , flat, $\beta=0$ ) | 43.6 [41.4, 45.8]        | 71.9 [68.8, 74.9]        | 56.8 [52.8, 60.8]        |

## 8 Ablation study

Table 4 removes one component at a time on all three pools at the  $4\times$  operating point, with paired intervals over records. The ordering matches the conference version’s single-dataset analysis, now replicated across datasets with intervals an order of magnitude tighter at  $n=1000$ ; the replication is itself a finding, because it shows the component structure is a property of the formulation rather than of one pool. Removing question conditioning costs a paired 16.3 points on QASPER [14.1, 18.3], 38.5 on HotpotQA [34.3, 42.5], and 23.1 on 2WikiMultihopQA [19.0, 27.5]—the single largest factor on every dataset, and its size tracks the strength of the question’s lexical signal: without the query the score rewards generally informative sentences, and evidence for a specific question is not the same as salient text, so the penalty is mildest on QASPER’s abstract questions and harshest on the entity-named multi-hop pools. This is the component-to-claim mapping for the multi-dataset result of Table 3: the margin over question-agnostic selectors is the measured value of the question, now measured three times. The redundancy penalty is recall-neutral on every dataset (-0.1 [-0.4, 0.3] on QASPER, comparable interval-straddling values on the other two pools): its work is diversity under tight budgets, refusing to spend scarce budget twice on near-duplicate sentences, which are common in the methods and related-work sections of scientific text. The recall protocol does not price that work—evidence recall is insensitive to whether the retained sentences repeat each other. We therefore measured the penalty’s work directly, on the same pools and budgets, with two selector-level metrics: the mean pairwise IDF-weighted cosine similarity of the selected set (the quantity the penalty targets) and the duplicate content-token rate of the emitted text (a token-level metric the selector does not optimize). Removing the penalty raises the selected set’s mean pairwise similarity by 0.04 of a percentage point [0.03, 0.05] on QASPER at the same  $4\times$  budget, by 0.20 [0.13, 0.28] on HotpotQA and 0.13 [0.07, 0.21] on 2WikiMultihopQA, and raises the duplicate content-token rate by 0.55 [0.49, 0.60] percentage points, with the same direction on the other two pools and at every budget from  $2\times$  to  $16\times$ ; the duplicate-rate effect grows as the budget tightens (from 0.31 points at  $2\times$  to 0.76 at  $16\times$  on QASPER), which is the diversity-under-tight-budgets

mechanism operating as designed. The paired emitted-word difference never exceeds a tenth of a word, so the comparison is budget-matched. The effect is statistically unambiguous but modest in absolute terms: 39.9% of the emitted content tokens are duplicates under the full selector versus 40.4% without the penalty on QASPER, and the redundancy-free relevance-ranking configurations sit above both—relevance-only and BM25 duplicate at 41.9% and 41.9%—the same cross-implementation agreement the recall layer shows. The penalty’s diversity is thus a measured property with a small absolute size, bought at its measured costs: recall neutrality on every dataset and a few milliseconds of selector arithmetic; we claim no recall benefit from it.

The two remaining rows are the table’s most instructive finding, and BM25 makes it legible from the outside as well as the inside, on every dataset. A relevance-only configuration (question conditioning with a flat prior and no penalty) reaches 43.6%, 71.9%, and 56.8% on the three pools—the highest number in every column—and BM25, an entirely external implementation of relevance ranking, lands within a point of it on every dataset (43.1%, 73.1%, 57.6%). The agreement is the point: the relevance term, implemented either way, is a reference ceiling for evidence *recovery* under 1, and everything Sieve adds on top of it is a deliberate trade of raw recall for other objectives. The positional prior behaves differently, and the difference is now measured to be dataset-stable: flattening it recovers a paired  $|-7.1|$  points on QASPER  $[-9.1, -5.1]$ ,  $|-6.0|$  on HotpotQA  $[-8.5, -3.8]$ , and  $|-6.4|$  on 2WikiMultihopQA  $[-9.7, -3.4]$ . The U shape actively deprioritizes middle sentences, all three pools keep their evidence near the middle (median relative evidence position 48%, 54%, 46%), and so the prior pays a comparable price everywhere. The same U shape, however, places retained evidence where readers attend most reliably, which is the trade-off [section 5](#) formalized as a reliability allocation and [section 10](#) measures position by position. The component ablations and the multi-dataset results compose into a single account of when Sieve wins: the margin over question-agnostic selection scales with the strength of the lexical signal (1 of 3), the prior converts part of it into placement, and both terms are visible in [Table 3](#) as the dataset-dependent gap over question-agnostic selection, and in the BM25 rows as the price of the prior. Read through the formulation of [section 3](#), the table says the same thing on every dataset from opposite sides of the selector: relevance is the objective that maximizes recall, positional reliability and diversity are the objectives that spend it, and the budget decides the exchange rate—which is why we report the pair as a structural trade-off between selection objectives ([Table 1](#)) rather than as a contest between two selectors.

## 8.1 Budget dependence of the component trade-offs

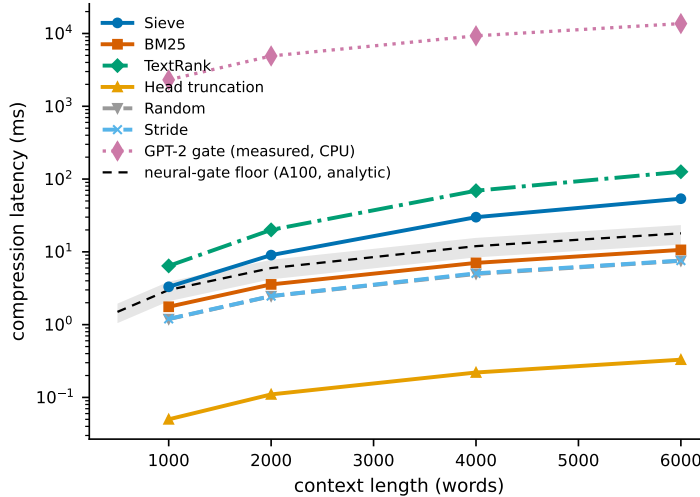
[Table 5](#) extends the ablation from the single  $4\times$  operating point to the full budget range on QASPER; [Figure 4](#) draws the same matrix on all three datasets as cost–recall frontiers with equal-budget lines. Three budget regularities emerge, none of them visible at a single operating point. First, the question’s value is not a constant of the budget: the question-agnostic family collapses as the budget tightens (truncation falls from 50.0% at  $r=2$  to 4.4% at  $r=16$ , TextRank from 51.3% to 7.5%), while the question-conditioned family degrades slowly, and the no-question variant of Sieve itself collapses from 43.3% to 3.7%—question conditioning is what keeps compression

**Table 5** Evidence recall (%) of every configuration at four budgets on the QASPER pool; each column is an equal-budget slice of the full configuration–budget–dataset matrix, which was measured on all three pools (bootstrap 95% intervals for the  $r=4$  column are given in Table 4; the full interval set ships with the released artifact, and Figure 4 draws all three pools). The question-agnostic family collapses as the budget tightens while the question-conditioned family degrades slowly; the prior’s price and the relevance-only/BM25 agreement persist at every budget.

| Configuration            | $r=2$ | $r=4$ | $r=8$ | $r=16$ |
|--------------------------|-------|-------|-------|--------|
| Sieve (full)             | 57.9  | 36.2  | 23.9  | 15.6   |
| BM25                     | 61.1  | 43.1  | 31.0  | 21.5   |
| no question conditioning | 43.3  | 19.9  | 8.1   | 3.7    |
| flat positional prior    | 62.8  | 43.3  | 31.6  | 22.0   |
| no redundancy penalty    | 57.6  | 36.3  | 24.1  | 15.8   |
| relevance only           | 61.1  | 43.6  | 31.1  | 21.5   |
| Head                     | 50.0  | 23.4  | 10.6  | 4.4    |
| Random                   | 50.1  | 26.0  | 13.4  | 6.6    |
| Stride                   | 41.6  | 22.2  | 12.0  | 5.9    |
| TextRank                 | 51.3  | 26.3  | 14.0  | 7.5    |

usable at tight budgets, and it is the mechanism, not the dataset, that decays. Second, the prior’s price is budget-stable: the flat-prior configuration leads the full one by five-to-eight points at every budget (62.8% against 57.9% at  $r=2$  and 22.0% against 15.6% at  $r=16$ ), the same six-to-seven paired points measured at  $r=4$  in Table 4. Third, the internal relevance-only configuration and external BM25 track each other within a point at every budget (61.1% against 61.1% at  $r=2$ , 21.5% against 21.5% at  $r=16$ ): the reference-ceiling agreement of Table 4 is not an artifact of one operating point.

A fourth regularity concerns the cost of the trade itself, and it is the one the formulation cares about most: the trade is computationally free. The five Sieve configurations differ only in scoring weights and selection coefficients, and their measured CPU latency at a 6,000-word context spans 50.9–54.3ms (subsection 9.1), a range within run-to-run noise of the same benchmark. Every point of recall the selector trades for placement and diversity is therefore paid in information units, not in milliseconds; the compute axis of the frontier separates selector *families* (truncation, statistical, neural), not configurations within a family, and the equal-budget lines of Figure 4 make this visible as the vertical spread of the Sieve cluster at constant cost. This is the empirical content of treating the three objectives as one selection problem: at fixed budget and fixed cost, the only degrees of freedom are where the objectives sit relative to each other, and the matrix measures exactly that.



**Fig. 3** Measured compression latency versus context length on the evaluation container (median of 5 runs; 3 for the gate; same box as every other measured number in this paper). The dashed band is the analytic wall-clock of one served forward pass of a 124M-parameter scorer over the full context on an A100 at 40% utilization, the minimum any perplexity-gated compressor must pay; coarse-to-fine pipelines repeat this pass at every budget iteration. The statistical selectors sit at or below the band with no model hosted; the measured CPU points for the GPT-2 gate (dotted) lie three orders of magnitude above it.

## 9 Efficiency analysis

This section reports the cost axis of the trade-off for every method, on identical hardware, and assembles the pieces into the frontier picture that the introduction advertised.

### 9.1 Compression latency

Figure 3 compares the measured compression cost of the statistical selectors (Sieve, BM25, TextRank, random, stride, and head truncation) across context lengths, and against the floor of perplexity-gated compression: one served forward pass of the GPT-2-scale scorer (Radford et al. 2019) over the full context. Sieve compresses a 6,000-word context in 53.7ms on the evaluation container’s cores, against 126.0ms for TextRank, 10.6ms for BM25, and 0.3ms for truncation; BM25 is the cheapest question-conditioned selector in the study, running at roughly 5× the speed of the full Sieve at the same length, a direct consequence of skipping the greedy marginal search. Every statistical selector’s cost is noise at serving scale: the end-task reader call is billed by tokens (1315 prompt tokens per compressed call against 5181 at full context), so the selector’s tens of milliseconds remove about three quarters of what the reader would otherwise prefill, and the projection of subsection 9.5 prices what the saved prefill and cache are worth at deployment scale.



**Table 6** The three-way trade-off at  $r = 4$  on QASPER, one row per method; recall is the selector protocol (evidence recall at  $4\times$ ,  $n=1000$ ), F1 the end-task protocol ( $n=100$ ), and latency the measured CPU cost of the compression itself on a 6,000-word context on the same container. The GPT-2 gate was not run under the end-task protocol; its GPU floor is analytic ( $2|M_s|L$  FLOPs at 40% utilization on an A100).

| Method       | extra model | GPU | recall $\uparrow$ | F1 $\uparrow$ | latency (CPU) | GPU floor            |
|--------------|-------------|-----|-------------------|---------------|---------------|----------------------|
| Full context | –           | no  | –                 | 33.6          | 0 ms          | –                    |
| Head         | –           | no  | 23.4              | <b>16.7</b>   | 0.3 ms        | –                    |
| Random       | –           | no  | 26.0              | 20.5          | 7.5 ms        | –                    |
| Stride       | –           | no  | 22.2              | 14.1          | 7.6 ms        | –                    |
| TextRank     | –           | no  | 26.3              | 18.4          | 126.0 ms      | –                    |
| BM25         | –           | no  | <b>43.1</b>       | 25.5          | 10.6 ms       | –                    |
| Sieve        | –           | no  | 36.2              | 20.7          | 53.7 ms       | –                    |
| GPT-2 gate   | 124M (int8) | yes | 28.7              | –             | 12.7 s        | 10.8 ms <sup>†</sup> |

<sup>†</sup>analytic, at the median context’s token count; the gate’s measured latency row is the 6,000-word operating point (mean over the 17 pool records at that length).

## 9.2 The neural gate, measured

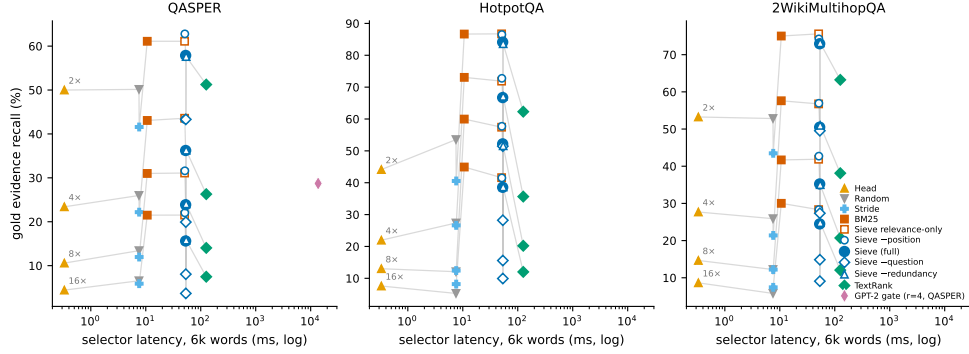
The conference version could only bound the neural compressor’s cost analytically; this version measures it. The GPT-2 perplexity gate of [subsection 6.3](#) runs on the same container, the same cores, and the same records as every other method. Three facts come out of the measurement. First, the gate’s scoring pass costs 8.2 s of CPU time on the median-context record (3591 words; median over records 7.5 s, 90th percentile 13.1 s), and 12.7 s on the 17 records near the 6,000-word cap (per-record means). On the timing benchmark of [Figure 3](#), which holds the context at exactly 6,000 words, the same pass costs 13.7 s against Sieve’s 53.7 ms: the  $255\times$  ratio reported in [Table 6](#), and  $320\times$  at the median-context operating point (where Sieve’s latency is interpolated to the same word count). Second, even granting the gate a GPU, the analytic A100 floor for the same pass is 10.8 ms at the median context’s token count (5410 tokens), at best comparable to Sieve’s entire CPU wall-clock; and the floor assumes the scorer is already resident, excludes tokenization and budget iterations, and prices the accelerator that Sieve does not have at zero. Third, the gate’s quality at  $4\times$  on QASPER is 28.7% evidence recall [20.4, 36.6] against Sieve’s 36.2% [33.9, 38.5]: on this benchmark the floor-cost configuration of the neural approach does not beat the statistical selector even on the reader-free metric, because GPT-2 perplexity is a proxy for predictability, not for question relevance: the gate keeps sentences the language model finds likely, which is not the same as sentences that answer the question. A stronger scorer (a fine-tuned reranker, or LLMingua-2’s distilled classifier) would be expected to close and pass Sieve on quality, at the same or larger hosted cost; the frontier statement of [subsection 9.4](#) covers both cases, and [section 13](#) states explicitly that the gate is the family’s cost floor, not its best quality.

### 9.3 The accuracy–compression–cost trade-off

Table 6 assembles the three axes. Reading it as the formulation of subsection 3.2 suggests: within the model-free rows, Sieve dominates Random, Stride, and TextRank on quality (separated recall intervals on all three datasets) at latency within two orders of magnitude of truncation; BM25 dominates Sieve on raw recall at a fifth of its latency, and the two are the complementary question-conditioned points whose trade the placement analysis prices; and no model-free method dominates the question-conditioned pair on any dataset at any ratio we measure. The gate row shows what the neural corner buys and costs at its floor configuration: its recall on QASPER is 28.7% against Sieve’s 36.2% (a deficit of 36.2–28.7 points with separated intervals; the gate also does not separate from head truncation at 23.4%), and its cost is a hosted 124M-parameter model, a matched tokenizer, and roughly 255× more measured CPU work at the 6,000-word operating point, or an accelerator whose analytic floor already matches Sieve’s entire CPU budget. The table is the paper’s answer to the question a practitioner actually asks: what does conditioning on the question cost, and what does a neural scorer buy over not hosting one? On these benchmarks the answer is: conditioning on the question is nearly free and worth tens of recall points over question-agnostic selection; relevance ranking alone (BM25) already captures most of that gain at the lowest question-aware latency; the floor-cost neural scorer, at this scale, does not even match truncation on recall, let alone buy back its own cost; and the quality that larger scorers would add is bought at hosted-model cost that the frontier of subsection 9.4 makes explicit.

### 9.4 The accuracy–efficiency frontier

Figure 4 plots the frontier directly, in the view the formulation suggests: quality against measured selector cost, with the compression budget as the parameter that moves a configuration along its equal-budget line, and with every configuration of subsection 8.1 drawn rather than a single operating point. The picture is the same on all three datasets. From truncation to question conditioning, quality is bought at almost no cost: every question-conditioned configuration—Sieve’s five variants and BM25—dominates every question-agnostic one on recall at every budget, at latencies within two orders of magnitude of free. From BM25 to the full Sieve, quality in the reader-free metric is knowingly traded for placement and diversity, and the trade is visible as the vertical drop from the relevance-only markers to the full configuration at the same abscissa: the objectives are traded at zero marginal compute (50.9–54.3 ms for all five variants, subsection 8.1). TextRank marks the far edge of the statistical family: it is the most expensive model-free selector measured here (126.0 ms against Sieve’s 53.7 ms at the 6,000-word operating point) while lying below the question-conditioned family on recall at every budget, so the question-conditioned statistical selectors dominate it on both axes. From the statistical selectors to the gate, a factor of roughly 255 in measured compute does not buy measurable quality on these benchmarks at this scorer scale; beyond the gate, larger and specialized scorers presumably buy quality again, at hosted-model cost. Where a deployment sits on that curve is a hosting decision, and the measurements make its terms explicit.



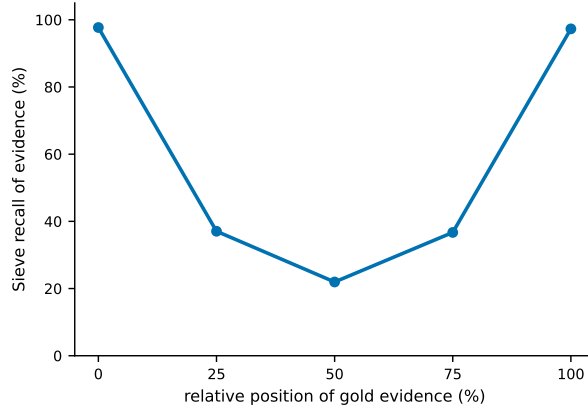
**Fig. 4** The accuracy–efficiency frontier of model-free selection and the measured gate, redrawn from the full configuration–budget–dataset matrix of [subsection 8.1](#): one panel per dataset, horizontal axis the measured CPU latency of each configuration at the 6,000-word operating point (log scale, same container), vertical axis gold evidence recall, thin gray lines connecting configurations at the same compression budget (2× to 16×). The five Sieve variants run at essentially identical cost (50.9–54.3 ms at 6,000 words), so the trade between raw recall and placement/diversity appears as the vertical spread of the Sieve markers at constant cost; BM25 is the cheapest question-conditioned point; TextRank is the most expensive model-free selector and is dominated on both axes by the Sieve family; the measured GPT-2 gate (QASPER,  $r=4$ ) sits three orders of magnitude to the right of Sieve at 28.7% recall. The question-conditioned statistical selectors occupy the low-cost, high-recall region at every budget and on every dataset.

**Table 7** Analytical projection at an 8k-token prompt and  $r = 4$ , computed from published LLaMA-2 architecture constants with the stated assumptions (bf16, A100 at 40% prefill utilization). Prefill latency and cache memory are the prefill-stage quantities.

| Model       | prefill (full) | prefill ( $r = 4$ ) | KV cache | cache ( $r = 4$ ) |
|-------------|----------------|---------------------|----------|-------------------|
| LLaMA-2 7B  | 0.86 s         | 0.22 s              | 4.2 GB   | 1.05 GB           |
| LLaMA-2 13B | 1.67 s         | 0.42 s              | 6.6 GB   | 1.64 GB           |
| LLaMA-2 70B | 8.8 s          | 2.2 s               | 2.6 GB   | 0.66 GB           |

## 9.5 Projected savings for LLaMA-class serving

[Table 7](#) translates a 4× compression into infrastructure terms for LLaMA-2 deployments ([Touvron et al. 2023](#)) using the standard prefill FLOPs approximation  $2N_{\text{params}}L$  for context length  $L$ , and cache memory of  $2n_{\text{layer}}n_{\text{kv}}d_{\text{head}}L$  elements in bf16, and an assumed 40% model-FLOPs utilization on one A100; the 70B row reflects grouped-query attention with eight KV heads. The projection is analytical, not measured, and the assumptions are stated so that a reader can recompute them; its role is to show that the recall results of [Table 3](#) buy cache and prefill savings that scale with the ratio rather than with the model, and that the compression’s own cost (53.7 ms on CPU at the 6,000-word operating point, 10.6 ms for BM25) is three orders of magnitude below the prefill it removes even for the smallest row.



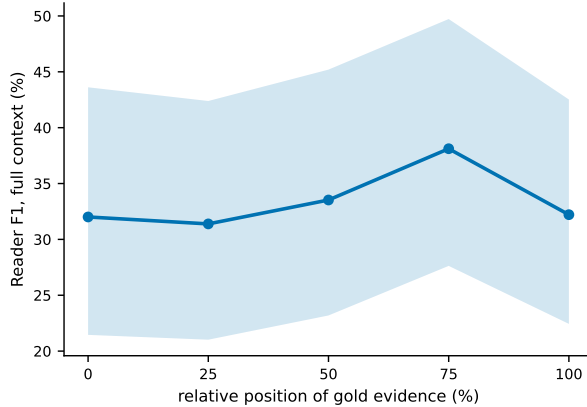
**Fig. 5** Selection-level recall of the gold evidence sentence as a function of its relative position among 40 unrelated filler sentences from the same paper (QASPER recall pool,  $n$  1000–1,000 per position; the placement protocol rebuilds contexts so nearly all pool records qualify). Sieve retains almost all evidence placed at the ends and a fifth of the evidence placed in the center: the U-shaped prior reproduces, at the selection level, the lost-in-the-middle profile documented for long-context readers (Liu et al. 2023), by design.

## 10 Robustness analysis

### 10.1 Evidence placement

Figure 5 shows the placement curve on the expanded pool. Retention is 97.7 at the start and 97.3 at the end, against 21.9 in the middle: the selector mirrors, at the selection level, the same lost-in-the-middle profile that readers exhibit (Liu et al. 2023), which is the design intent; at  $\alpha = 0.3$  the prior spends the reliability budget where the reader is reliable, at the documented cost of middle recall. Two consequences follow for practice. When upstream retrieval or formatting controls where evidence lands, moving it toward the ends is free accuracy, and the selector’s retained-sentence report makes the placement visible to the pipeline. When evidence placement cannot be controlled and middle recall matters more than reader-side reliability, the floor hyperparameter flattens the prior, and Table 4 quantifies exactly what that choice costs and buys, on every dataset: flattening recovers the middle for a dataset-stable six-to-seven paired points—43.3% total recall versus 36.2% on QASPER, 72.8% versus 66.7% on HotpotQA, 56.9% versus 50.5% on 2WikiMultihopQA—with the recovered evidence placed where readers use it least reliably. The deployment guidance is symmetric and explicit. BM25 occupies the opposite pole of this exchange: its retained evidence is placed exactly where relevance puts it, including the middle, which is why the placement axis, not the recall axis alone, is the right way to compare the two.

Figure 6 re-measures the reader side under the same reader and protocol family as Figure 5: answer F1 is 32.0 at the start, 33.5 in the middle, and 32.2 at the end (mid-minus-ends gap 1.4 F1 points, CI [-0.7, 3.7]). At this context scale—synthetic contexts of median  $\sim 995$  words, the regime where lost-in-the-middle is documented to weaken—the reader-side profile is flat, so the U-shaped prior remains a selector-level



**Fig. 6** Reader-side answer F1 (full context, no selection) with the gold evidence sentence placed at five relative positions; reader GLM-4-Plus,  $n=50$  per position, error bars are bootstrap 95% intervals. The profile is flat at this context scale: the mid-minus-ends gap is 1.4 F1 points (CI [-0.7, 3.7]).

**Table 8** Sensitivity of evidence recall at  $r = 4$  to the fixed hyperparameters on the QASPER recall pool ( $n = 1000$ ), one knob at a time. Recall is flat in  $\beta$  and mildly sensitive to the prior’s shape; it rises monotonically with floor, which is the recall–reliability trade-off of [section 8](#) seen from the other side.

|          |      |      |      |
|----------|------|------|------|
| $\beta$  | 0.0  | 0.3  | 0.6  |
| recall   | 36.3 | 36.2 | 36.3 |
| floor    | 0.0  | 0.35 | 0.7  |
| recall   | 32.1 | 36.2 | 41.2 |
| $\gamma$ | 0.5  | 1.5  | 3.0  |
| recall   | 39.0 | 36.2 | 36.9 |

design choice justified by the cited longer-context evidence ([Liu et al. 2023](#)), with its cost priced by [Table 4](#).

## 10.2 Hyperparameter sensitivity

[Table 8](#) reports the sensitivity grid on the expanded pool. Recall is flat in  $\beta$ , confirming that the redundancy penalty neither helps nor hurts evidence recovery at this ratio; its work is diversity, and [section 8](#) shows the same from the ablation side. [section 8](#) also measures that work directly, as duplicate-rate and pairwise-similarity reductions at matched budgets; the flat recall row above is the price side of the same trade. Recall

**Table 9** Evidence placement and question-style statistics per recall pool, and the resulting margins over head truncation and over BM25 at  $r = 4$ . Head truncation tracks the  $1/r=25\%$  prediction of 4 on all three pools (evidence placement is close to uniform), while the margins vary with how strongly the question’s terms identify the evidence sentences.

|                                 | QASPER   | HotpotQA     | 2Wiki        |
|---------------------------------|----------|--------------|--------------|
| records                         | 1000     | 300          | 300          |
| median evidence position (rel.) | 48%      | 54%          | 46%          |
| evidence sentences per record   | 4.8      | 2.5          | 2.4          |
| head recall at $4\times$        | 23.4     | 22.0         | 27.7         |
| Sieve recall at $4\times$       | 36.2     | 66.7         | 50.5         |
| BM25 recall at $4\times$        | 43.1     | 73.1         | 57.6         |
| margin over head (paired)       | 12.8     | 44.8         | 22.8         |
| margin vs BM25 (paired)         | -6.8     | -6.3         | -7.1         |
| question style                  | abstract | entity-named | entity-named |

risers as floor increases, the quantitative statement of the placement trade-off: a flatter prior keeps middle-placed evidence at the cost of reader-side reliability. The prior’s sharpness  $\gamma$  moves recall by a few points without a clear trend. None of the three knobs changes the qualitative conclusions of Table 3 and Table 4: at every setting the question-conditional term dominates, and the ordering of methods is unchanged.

### 10.3 Cross-dataset transfer without retuning

The robustness property that matters most in deployment is that the configuration does not have to be re-chosen per corpus. All results in Table 3 use the configuration tuned once on the 20-record development split—ten QASPER records disjoint from the test pool plus ten HotpotQA distractor records—and re-confirmed by the expanded dev grid (Appendix C):  $\alpha = 0.3$ ,  $\beta = 0.3$ , floor = 0.35,  $\gamma = 1.5$ , applied unchanged to QASPER at  $n=1000$ , HotpotQA at  $n=300$ , and 2WikiMultihopQA at  $n=300$ , including the two datasets that entered the study after the configuration was frozen. The margins over every question-agnostic baseline hold on all three, which is the transfer claim; the margin’s size varies with the question style, which section 12 attributes to the lexical signal rather than to any per-dataset adaptation. A deployer who cannot hold out tuning data inherits the same behavior: the sensitivity table shows the method degrades gracefully, not cliff-like, when every knob is mis-set by its grid step.

## 11 How loosely recall bounds accuracy

The paper’s two quality protocols measure different things, and the difference is a result worth reporting rather than hiding. We join the two at the sample level: for every record of the end-task test split that carries locatable evidence (100 of the 100), and every selector that ran under both protocols, we pair the record’s evidence recall

at  $4\times$  with its end-task F1 at  $4\times$ , and compute the correlation over the pooled 600 method-record pairs.

The correlation is positive and weak: pooled Pearson 0.22, Spearman 0.21; per method it ranges from 0.04 (Sieve) to 0.41 (head truncation), with TextRank at 0.32, and every per-method coefficient is positive—weak but sign-stable at  $4\times$  the conference sample, so the two protocols remain separate instruments even where both are available. The direction is as expected (records whose evidence survives compression are answered better), but the tightness is not: a reader that fails a fully-evidenced record, or answers correctly from a partially-evidenced one, contributes noise that a selector-level metric cannot see, and the per-method coefficients are individually imprecise at this scale. Two conclusions follow. First, evidence recall is a necessary-but-not sufficient condition for end-task quality: a record whose evidence is discarded cannot be answered from the compressed context, but a fully-evidenced record can still be answered wrongly, and at this sample size the association between the two is weak—evidence recall is an incomplete proxy for downstream quality, not an upper bound on it. Second, and methodologically, weak pooled correlation does not license reading recall gains as accuracy gains in either direction: it neither predicts nor refutes individual method-level orderings, as shown by the resolved end-task gap of  $|-4.8|$  paired F1 points, by which BM25’s selector-level margin transferred end to end—one confirmed transfer among selectors whose sample-level coupling remains loose; which is why the two protocols are reported separately throughout this paper.

## 12 Discussion

### *When is model-free selection enough?*

The measurements suggest a practical rule. If the query’s content terms recur in the passages that answer it (entity-named questions, retrieval over technical documentation, FAQ traffic), the statistical selector captures most of the recoverable evidence at milliseconds of cost, and a hosted scorer has little left to buy. If the link between question and evidence is semantic rather than lexical (paraphrase-heavy corpora, cross-lingual traffic, questions about implication), [1](#) fails by construction, and the neural corner earns its infrastructure. The corpus statistic of [section 12](#) (how often question terms recur in evidence) estimates which regime a deployment is in from a labeled sample of a few hundred records, and the recall protocol itself is cheap enough to run on that sample before committing.

### *Relevance ranking versus budgeted selection: the BM25 lesson.*

The most instructive baseline in the study is also its strongest on raw recall: BM25 does nearly everything Sieve’s relevance term does, at a fifth of the latency, and it beats the full selector on raw recall on every dataset. A reader of [Table 3](#) alone would conclude that BM25 is the better compressor, and the honest answer is that for *pure evidence recovery at generous budgets* it is. What Sieve adds is a budgeted selection policy on top of relevance: evidence placed where readers attend reliably, near-duplicates refused under tight budgets, and an explicit knob (floor) pricing the exchange. The end-task protocol at  $n=100$  separates the two, in BM25’s favor, with a paired interval of  $[-8.6,$

-1.4] F1 points (retain 76.0% against 61.8%), so the recall ordering transferred at this operating point. The measured reading is conditional: for deployments optimizing end-task F1 alone at this operating point on QASPER-like traffic, the measured ordering favors BM25. The practical reading keeps its conditional shape, with the end-task price now measured: for deployments that feed a fixed-prefix-friendly reader or that value recall above all, the measurements favor BM25 (or the relevance-only ablation, at the same recall); for deployments whose readers exhibit positional bias, or that operate at tight budgets where duplicates crowd out evidence, the full selector is the configuration that matches those requirements, and the measured end-task price of that choice at this operating point is  $|-4.8|$  paired F1 points. Either way, both points cost milliseconds and no model, and both dominate the question-agnostic corner that production pipelines still default to.

### *The frontier as a decision tool.*

The point of reporting all three axes for every method is that compression decisions are usually made on one. A team that optimizes quality alone will host a scorer and pay for it on every call; a team that optimizes cost alone will truncate and lose tens of recall points on exactly the traffic where questions carry signal; Table 6 prices the middle of that spectrum. The placement of the statistical selectors at the low-cost end of the measured frontier is the empirical content of the paper: the first two orders of magnitude of selector budget buy question conditioning, and the next two, spent on the gate’s scorer configuration, buy nothing measurable on these benchmarks.

### *Composition with serving-stack work.*

Sieve compresses the submitted context; attention and cache work compresses what the model does with it. The two multiply: a  $4\times$  shorter prompt enters FlashAttention (Dao et al. 2022) with a quarter of the prefill FLOPs and paged KV memory (Kwon et al. 2023) with a quarter of the pages, before StreamingLLM or H2O policies run on what remains (Xiao et al. 2023; Zhang et al. 2023). Nothing in the selector conflicts with those layers, and the audit trail (which sentences survived, with their scores) composes with the cache policies’ own bookkeeping.

### *Auditability.*

A selection rule whose every score is a countable statistic has a property the neural corner does not provide at zero configuration: any retained or discarded sentence can be explained to a stakeholder in one sentence of arithmetic. Whether that property matters is deployment-specific; it is a potential advantage for pipelines where compression decisions face audit requirements—a surface that regulated domains such as legal discovery or clinical records can require—and Sieve produces the explanation as a by-product of selection rather than through a post-hoc explainer. We have not evaluated any such deployment and claim no measured result here; the property is architectural, not empirical.



***Why the margin varies by dataset: the lexical signal.***

Table 9 assembles the mechanism behind the multi-dataset result. Evidence placement is close to uniform on all three pools (median relative position 48, 54, and 46%), so 4 predicts head recall near  $1/r = 25\%$ , and the measured values (23.4, 22.0, 27.7) sit within a few points of the prediction on every dataset. What varies is the strength of the lexical link between question and evidence. HotpotQA and 2WikiMultihopQA questions are compositional over named entities: the question names the film, the person, the year, and the evidence sentences contain exactly those tokens, so IDF-weighted coverage fires at full strength and 1 holds often. QASPER questions ask what, how, and why about a paper’s method: the query terms are common words whose evidence sentences need not repeat them, so the condition holds less often and the margins shrink from 44.8 and 22.8 points over truncation to 12.8. The BM25 margins vary in the same direction (-6.3 on HotpotQA, -6.8 on QASPER), because BM25 rides the same lexical signal. The generalization claim is therefore precise rather than blanket: the ordering of methods (question-conditioned above question-agnostic at every ratio beyond two) transfers to all three datasets, the mechanism (lexical coverage, per 3) is constant, and the margin scales with a measurable property of the corpus: how much of the question’s vocabulary recurs in its evidence. A practitioner can estimate that property on a sample of their own traffic before deciding between a statistical selector and a hosted scorer.

***Where the margin lives: the lexical regime.***

The dataset ordering also rules out a QASPER-specific reading of Sieve. The two multi-hop corpora are the ones where a reader without any compression would be asked to integrate evidence from two short paragraphs scattered among eight distractors; the selector’s job there is exactly lexical triage, and Sieve answers it with recall 66.7 and 50.5 at  $4\times$ . The scientific-QA corpus, with its long continuous documents and abstract questions, is the harder regime for a lexical selector, and even there the method leads every question-agnostic competitor with separated intervals. Between the two regimes, the ratio sweep of Figure 2 shows the same shape on all three datasets, which supports reading Sieve as a first stage for evidence-annotated, retrieval-heavy QA settings—with the lexical-signal boundary of section 12 as its explicit scope condition—rather than a benchmark specialist.

## 13 Limitations

Sieve inherits the reach of lexical statistics, and the theory section states the condition rather than hiding it: where evidence shares no vocabulary with the question, 1 fails and the relevance term goes blind. Paraphrased evidence scores poorly; multilingual inputs need language-matched stopword lists and segmentation; numbers, citations, and code identifiers, which dominate scientific text, are weighted only crudely by IDF. The margin analysis of section 12 makes the dependence measurable, not mysterious, and the BM25 rows show the same dependence from the outside.

The evaluation has five boundaries. First, the end-task protocol covers one reader (GLM-4-Plus) on one benchmark (QASPER) at one operating point ( $4\times$ ) with  $n=100$

across seven selectors including BM25: its paired half-widths (6.3 F1 points against truncation) cannot yet separate the question-agnostic orderings, the resolved BM25 gap (half-width 3.6 points) is single-benchmark, single-reader evidence, and reader-dependence is untested. Widening to multiple readers and cross-dataset end-task protocols remains the next step, and the released, checkpointed protocol scripts cover it without touching any selector-level claim. Second, the neural comparison instantiates the perplexity-gate mechanism with GPT-2 small at sentence granularity; it is the family’s cost floor and a faithful mechanism replica, but it is not LLMingua’s full coarse-to-fine machinery and not LLMingua-2’s distilled classifier, so the quality row of Table 6 is a lower bound on what the neural corner can reach, and the cost row a lower bound as well. Third, the reader-side positional profile was re-measured under our own reader at  $n=50$  paired records: at this context scale (median  $\sim 1,000$  words) the profile is flat—the mid-minus-ends gap is 1.4 F1 points with CI  $[-0.7, 3.7]$ —so the U prior remains a selector-level design choice justified by cited longer-context evidence (Liu et al. 2023), and its cost is priced by the ablation of Table 4. Fourth, the recall-accuracy correlation of section 11 is estimated at  $n=100$  per method (600 pooled pairs), so its magnitude remains imprecise even though every per-method coefficient is positive at  $4\times$  the conference sample. Fifth, sentence-level granularity bounds the useful compression range: beyond  $r \approx 16$  the output degenerates toward a few isolated sentences, and token-level compressors retain an advantage at extreme ratios.

A further boundary concerns the comparison landscape rather than our measurements: QASPC (Chen et al. 2025), the closest published relative in stated properties (training-free, question-aware, sentence-level with clause-level redundancy reduction), was not accessible in full text at the time of writing, so no measured comparison against it is reported and the design-space table records its stated properties only. Replicating the end-task protocol across additional readers and datasets is the remaining gap between the selector-level evidence and the reader-level claims; the released, checkpointed protocol scripts cover it without touching any selector-level claim, and any newly accessible competitor can be placed on the same measured frontier by the same scripts.

Two accounting caveats remain from the conference version. The reader experiments measure accuracy under a fixed serving stack rather than wall-clock serving of an open model, and the LLaMA-2 projection is analytical with stated assumptions rather than a measured deployment. The gate’s GPU floor is analytic and assumes a resident scorer at 40% utilization; a hosted implementation with tokenization, budget iterations, and cold starts costs more, and our CPU measurement of the gate is correspondingly the like-for-like number for CPU-bound deployments.

## 14 Conclusion

We asked how much of long-context quality a compressor can keep with no neural scorer at all, and answered it as a frontier study rather than a single operating point. At the  $4\times$  compression, Sieve keeps 61.8% of full-context F1 end to end, statistically matched with truncation and the question-agnostic floor at  $n=100$  and  $|-4.8|$  paired

F1 points below the relevance-only BM25 reference (76.0% retained)—the same ordering its selector-level recall margin predicts—while spending 53.7 ms of CPU time and zero model parameters. Under the reader-free protocol, on 1600 evidence-annotated records across three benchmarks, it retains more gold evidence than every question-agnostic selector with fully separated intervals, and the margin over truncation grows with the compression ratio and with the strength of the question’s lexical signal, from 12.8 points on abstract scientific questions to 44.8 on entity-named multi-hop ones. The added baselines price the remaining design space: BM25 shows that raw relevance ranking already recovers the most evidence at the least question-conditioned cost, which the selector’s own ablations explain: the full configuration knowingly trades that recall for positional reliability, and the measured GPT-2 perplexity-gate configuration, the cost floor of that family, does not exceed the statistical selector on these benchmarks at  $255\times$  the measured compute (6,000-word operating point) and a hosted model. The formal analysis states when a stronger scorer should be expected to earn its infrastructure: exactly when the question–evidence link stops being lexical. We see the paper’s legacy as threefold: a formulation of context compression as budgeted question-conditioned information selection; a measured three-axis frontier on which any selector—statistical, lexical, or neural—can be priced before justifying its infrastructure; and Sieve, the no-hosted-model instantiation of that formulation, usable as the low-cost first stage of a compression stack (composable with cache-level methods, auditable by construction, and tuned once and transferred).

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- **Competing interests:** The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.
- **Author contribution: Chang Tan:** Conceptualization, Methodology, Software, Investigation, Data curation, Writing – original draft, Writing – review & editing.
- **Data availability:** The benchmarks are public (QASPER, HotpotQA, 2Wiki-MultihopQA). The released repository contains the pool-construction scripts with fixed seeds, the sampled pools, every raw checkpoint, and one command per table and figure, so that every number in this paper can be regenerated from scratch.

## Appendix A Implementation details

Sentence segmentation splits on punctuation boundaries with guards for common abbreviations (*et al.*, *e.g.*, *Fig.*) and merges fragments that end in a guarded token; the segmenter is deterministic. Tokenization lowercases and keeps alphanumeric and hyphenated terms; stopwords are a fixed 150-word English list included with the code. The conference-version experiments ran on one core of a container with 8 vCPUs and 32 GB of memory; the journal-version experiments, including the GPT-2 gate (int8, two threads) and every timing number in [section 9](#), ran in a container with 2

vCPUs and 4 GB of memory, so that all same-hardware comparisons are like-for-like. Timing figures are medians of five runs after one warm-up (three for the gate). The full pipeline (segmentation, scoring, selection) is a single Python module with NumPy as its only dependency beyond the standard library; the gate baseline additionally uses Transformers and PyTorch. The repository reproduces every table and figure from the raw checkpoints with one command per artifact, and the macro file that carries every number in this paper into its LaTeX source is itself generated from those checkpoints, so no number in the text was typed by hand.

## Appendix B Proofs and formal statements

*Proof of 1* Each pipeline stage touches each token a constant number of times: segmentation and tokenization are single passes over the  $W$  words; the IDF document-frequency table is one accumulation pass over sentence term sets; relevance evaluates each sentence against the query term set in  $O(|s_i|+|Q|)$ ; the prior is  $O(n)$ . The greedy loop performs at most  $k$  iterations, each scanning the  $O(n)$  remaining candidates and evaluating the marginal penalty against the  $O(k)$  selected sentences at  $O(\bar{s})$  per pair on sparse counters, giving  $O(k n \bar{s})$  worst case; peak memory stays at the token counters because the released implementation recomputes similarities rather than memoizing them.  $\square$

*Proof of 2* A forward pass of a decoder-only transformer with  $N$  parameters over  $L$  tokens costs approximately  $2NL$  FLOPs for the matrix multiplications that dominate the arithmetic (Kaplan et al. 2020); any scorer-based compressor computes at least one such pass over the full context before it can rank its units, and coarse-to-fine budget searches compute one per iteration. The scorer must be resident, so  $c_{\text{host}} \neq 0$ .  $\square$

*Proof of 3* Greedy selection under  $\alpha = 1$ , flat prior, and  $\beta = 0$  orders sentences by  $\text{rel}$  alone. Under 1, for any  $e \in E$  and  $s \in C \setminus E$ ,  $\text{rel}(e) \geq \text{rel}(s)$ , so every evidence sentence is admitted before every dominated non-evidence sentence; ties among evidence sentences do not affect the claim because the budget condition  $B \geq \sum_{e \in E} |e|$  admits all of  $E$  whenever they are reached, and they are reached before the budget can be exhausted by non-evidence sentences that follow them in the order. Hence  $\rho = 1$ .  $\square$

*Proof of 4* Truncation keeps the prefix of  $\lceil n/r \rceil$  sentences. For an evidence sentence at a uniform position, the retention probability is  $\lceil n/r \rceil / n$ , which lies in the interval  $[1/r, 1/r + 1/n]$ , and independence across evidence sentences plus linearity of expectation gives  $\mathbb{E}[\rho] = \min(1, 1/r) + O(1/n)$ .  $\square$

## Appendix C Additional results

### *Compute accounting.*

The reader evaluation consumed 1010 completed API calls: the  $n=100$  seven-method main comparison, the development grid, and the reader-side positional profile of Figure 6 (50 paired records at five positions). The selector-protocol runs covered 1600 records across three pools with roughly forty selector invocations per record (methods, sweeps, ablations, sensitivity); the GPT-2 gate ran 100 QASPER records at 7.5 s

median scoring each. Compression-side compute for all journal experiments was under 6 vCPU-hours including the gate; no GPUs were used for any measurement reported in this paper.

### *Development grid.*

The development grid over  $\alpha \in \{0.3, 0.6, 0.9\}$  at  $r = 4$  gave dev F1 of 39.6, 38.9, and 39.0 respectively; the differences are within the noise of a 20-example development split, and we selected  $\alpha = 0.3$  for the best dev F1. The dev and test numbers are not directly comparable: the development split mixes ten QASPER records with ten HotpotQA distractor records, and the multi-paragraph HotpotQA items are far easier for the reader, which inflates the dev mean relative to the QASPER-only test split. The  $\alpha$  choice is a point estimate from a noisy grid, and the paper’s claims do not rest on it being optimal.

### *Conference-version compatibility.*

The conference version’s QASPER pool (191 records), its timing host, and its numbers remain reproducible from the original artifacts; the journal pools are supersets drawn under the same filters, and every qualitative conference claim (ordering of methods, growth of the margin in  $r$ , component attribution) reproduces at the larger scale with tighter intervals. Three conference-version statements are superseded: the QASPER recall intervals, which touched at their edges at the conference sample ( $n = 191$ ), are separated at the journal sample ( $n=1000$ ); the neural-baseline comparison, which was analytic in the conference version, is measured here; and the baseline set now includes BM25, which the conference version lacked entirely.

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